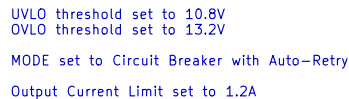


	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch			
B						
C						
D						
	1	2	3	4	5	

Sheet: /		
File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 1/21

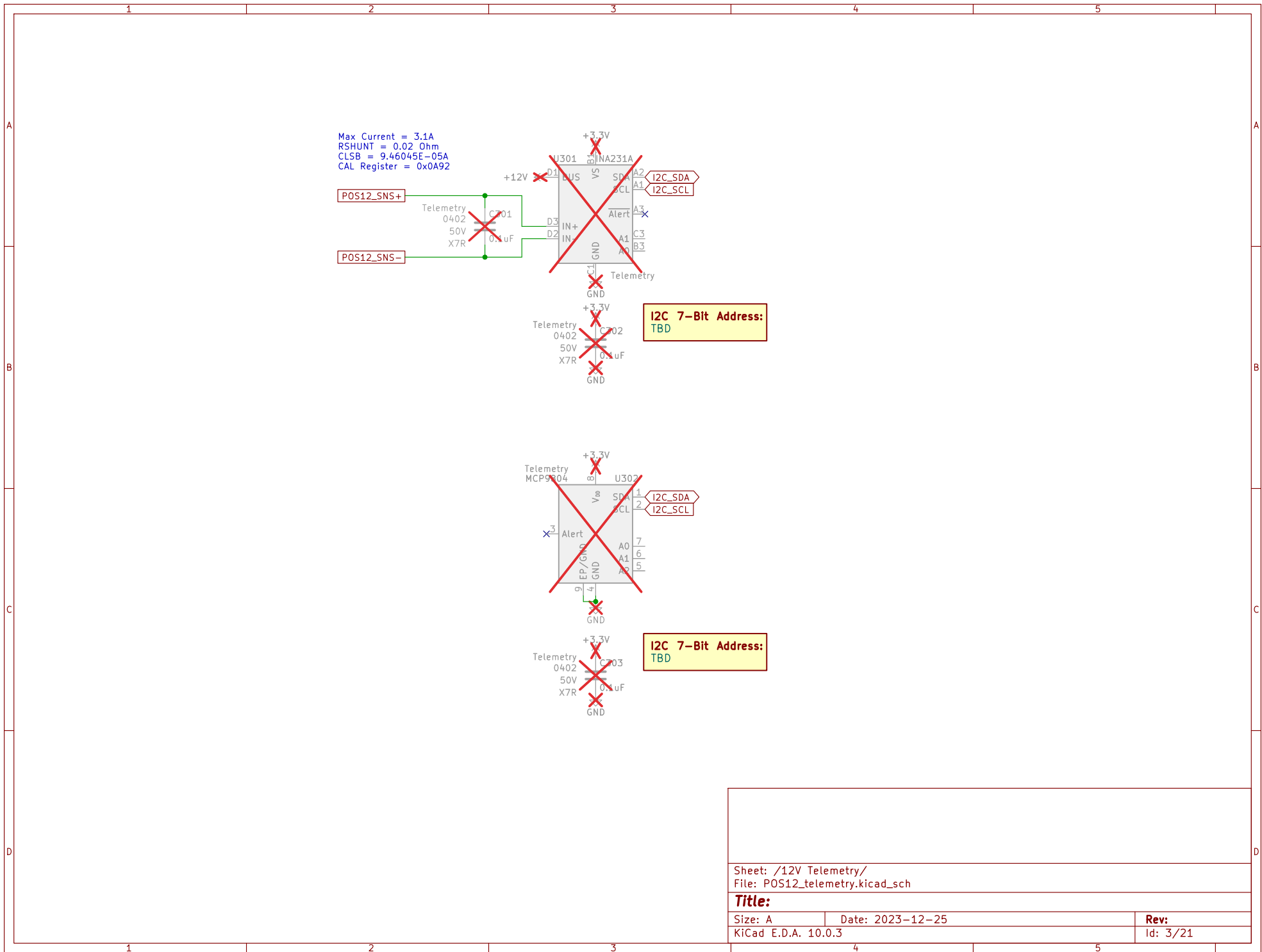
Explore shunt drop/dissipation at rated current



Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

Size: A	Date: 2023-12-25
KiCad E.D.A. 10.0.3	

Rev:
Id: 2/21



12V

3.3V

TP401

PWR_FLAG

3.3V

SHRINK

U402

TP562130A

SW

VOS

PG

SS/TR

DEF

FSW

AGND

EP

PGND

POS3P3_PG00D

U401

74LVC1G97

POS3P3_RUN

R401

R402

C401

C402

C403

C404

C405

C406

C407

C408

R403

R404

R405

R406

R407

R409

R410

Q401

Q402

Q403

Q404

Q405

Q406

Q407

Q408

Q409

Q410

Q411

Q412

Q413

Q414

Q415

Q416

Q417

Q418

Q419

Q420

Q421

Q422

Q423

Q424

Q425

Q426

Q427

Q428

Q429

Q430

Q431

Q432

Q433

Q434

Q435

Q436

Q437

Q438

Q439

Q440

Q441

Q442

Q443

Q444

Q445

Q446

Q447

Q448

Q449

Q450

Q451

Q452

Q453

Q454

Q455

Q456

Q457

Q458

Q459

Q460

Q461

Q462

Q463

Q464

Q465

Q466

Q467

Q468

Q469

Q470

Q471

Q472

Q473

Q474

Q475

Q476

Q477

Q478

Q479

Q480

Q481

Q482

Q483

Q484

Q485

Q486

Q487

Q488

Q489

Q490

Q491

Q492

Q493

Q494

Q495

Q496

Q497

Q498

Q499

Q500

Q501

Q502

Q503

Q504

Q505

Q506

Q507

Q508

Q509

Q510

Q511

Q512

Q513

Q514

Q515

Q516

Q517

Q518

Q519

Q520

Q521

Q522

Q523

Q524

Q525

Q526

Q527

Q528

Q529

Q530

Q531

Q532

Q533

Q534

Q535

Q536

Q537

Q538

Q539

Q540

Q541

Q542

Q543

Q544

Q545

Q546

Q547

Q548

Q549

Q550

Q551

Q552

Q553

Q554

Q555

Q556

Q557

Q558

Q559

Q560

Q561

Q562

Q563

Q564

Q565

Q566

Q567

Q568

Q569

Q570

Q571

Q572

Q573

Q574

Q575

Q576

Q577

Q578

Q579

Q580

Q581

Q582

Q583

Q584

Q585

Q586

Q587

Q588

Q589

Q590

Q591

Q592

Q593

Q594

Q595

Q596

Q597

Q598

Q599

Q600

Q601

Q602

Q603

Q604

Q605

Q606

Q607

Q608

Q609

Q610

Q611

Q612

Q613

Q614

Q615

Q616

Q617

Q618

Q619

Q620

Q621

Q622

Q623

Q624

Q625

Q626

Q627

Q628

Q629

Q630

Q631

Q632

Q633

Q634

Q635

Q636

Q637

Q638

Q639

Q640

Q641

Q642

Q643

Q644

Q645

Q646

Q647

Q648

Q649

Q650

Q651

Q652

Q653

Q654

Q655

Q656

Q657

Q658

Q659

Q660

Q661

Q662

Q663

Q664

Q665

Q666

Q667

Q668

Q669

Q670

Q671

Q672

Q673

Q674

Q675

Q676

Q677

Q678

Q679

Q680

Q681

Q682

Q683

Q684

Q685

Q686

Q687

Q688

Q689

Q690

Q691

Q692

Q693

Q694

Q695

Q696

Q697

Q698

Q699

Q700

Q701

Q702

Q703

Q704

Q705

Q706

Q707

Q708

Q709

Q710

Q711

Q712

Q713

Q714

Q715

Q716

Q717

Q718

Q719

Q720

Q721

Q722

Q723

Q724

Q725

Q726

Q727

Q728

Q729

Q730

Q731

Q732

Q733

Q734

Q735

Q736

Q737

Q738

Q739

Q740

Q741

Q742

Q743

Q744

Q745

Q746

Q747

Q748

Q749

Q750

Q751

Q752

Q753

Q754

Q755

Q756

Q757

Q758

Q759

Q760

Q761

Q762

Q763

Q764

Q765

Q766

Q767

Q768

Q769

Q770

Q771

Q772

Q773

Q774

Q775

Q776

Q777

Q778

Q779

Q780

Q781

Q782

Q783

Q784

Q785

Q786

Q787

Q788

Q789

Q790

Q791

Q792

Q793

Q794

Q795

Q796

Q797

Q798

Q799

Q800

Q801

Q802

Q803

Q804

Q805

Q806

Q807

Q808

Q809

Q810

Q811

Q812

Q813

Q814

Q815

Q816

Q817

Q818

Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	4	4.4mA
MCP6024	5.4mA	2	10.8mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	2	0.4mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder	50mA	1	50mA
Module	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.554W	168mA

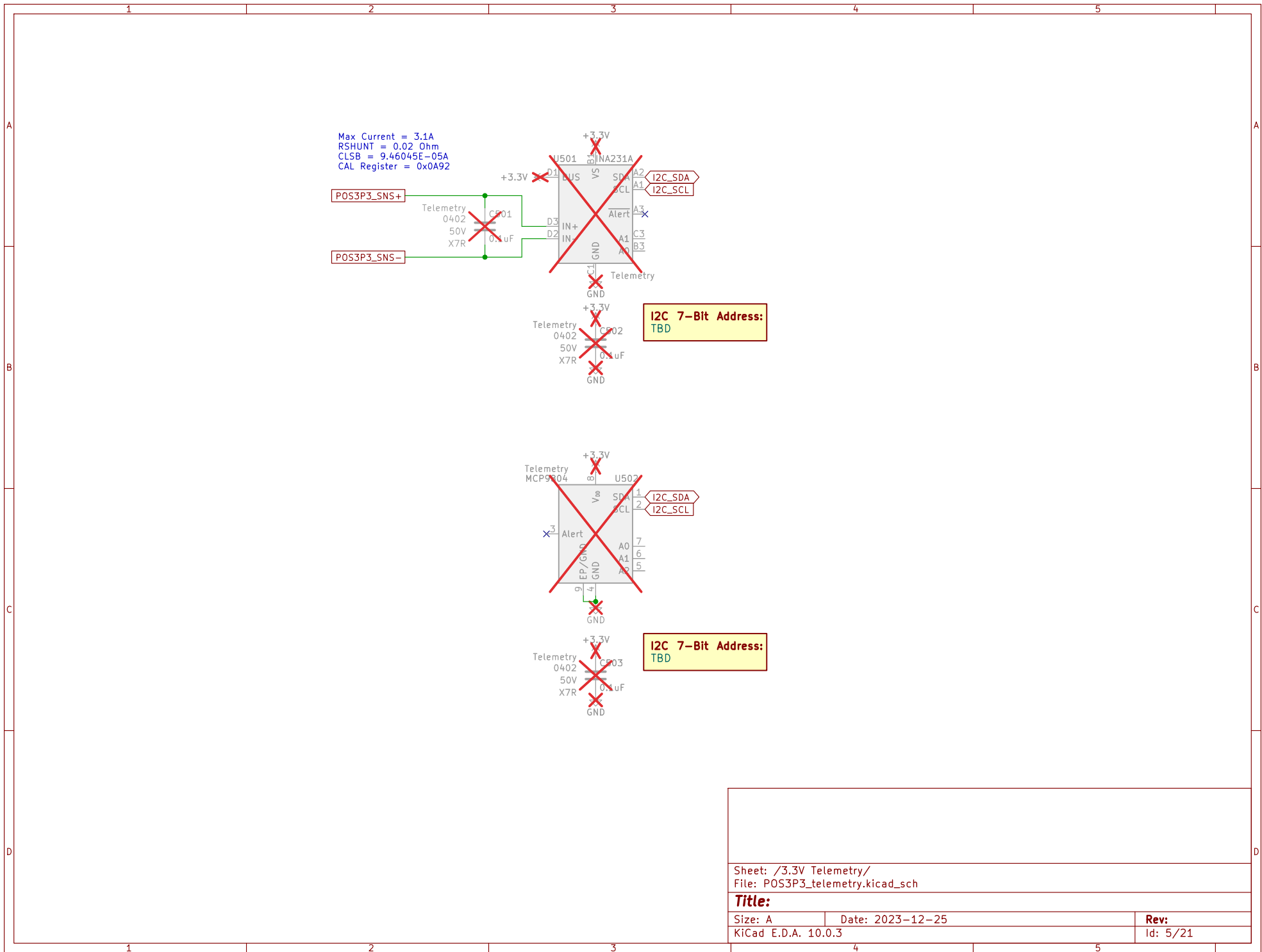
Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

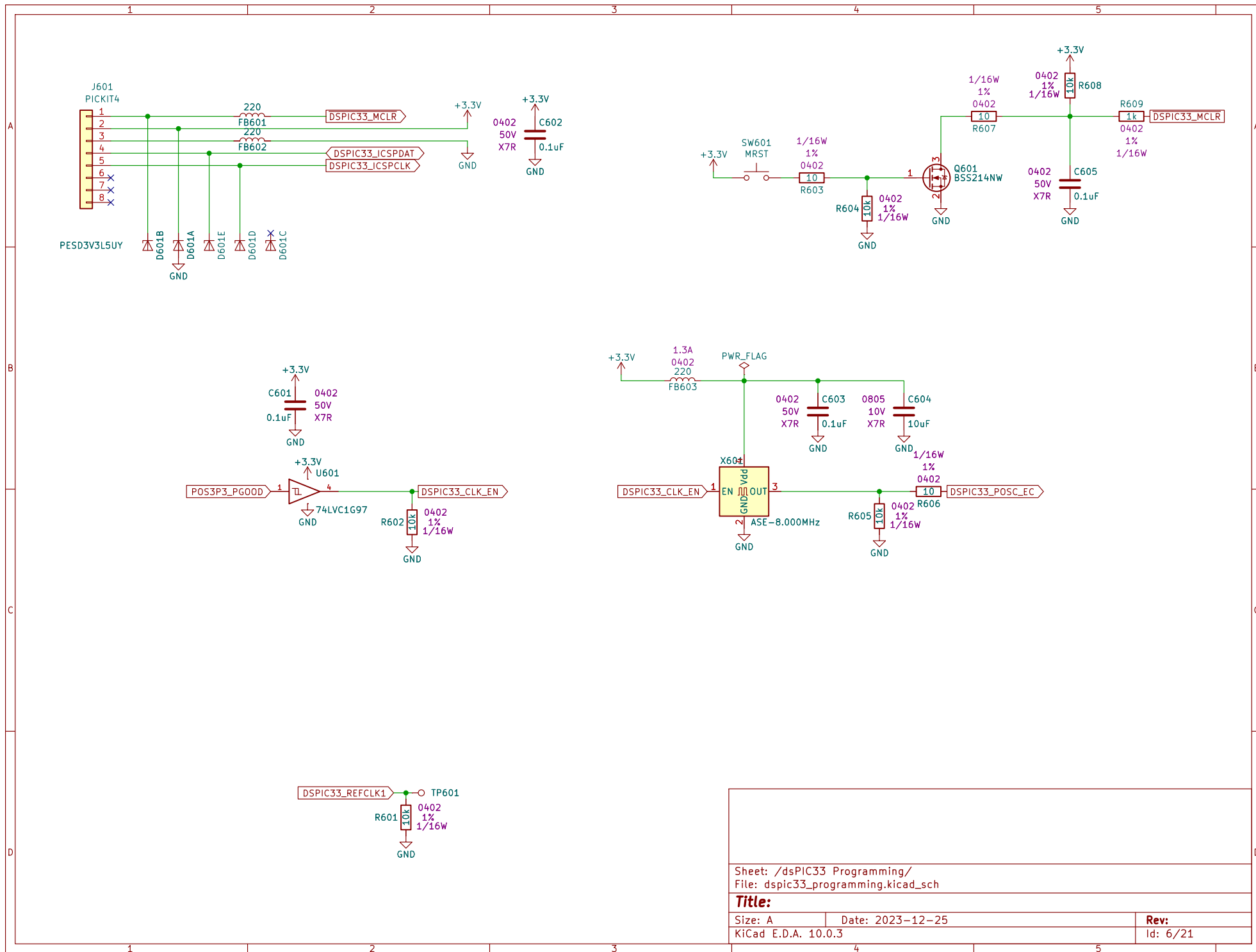
Title:

Size: A	Date: 2023-12-25
---------	------------------

KiCad E.D.A. 10.0.3

Rev:
Id: 4/21





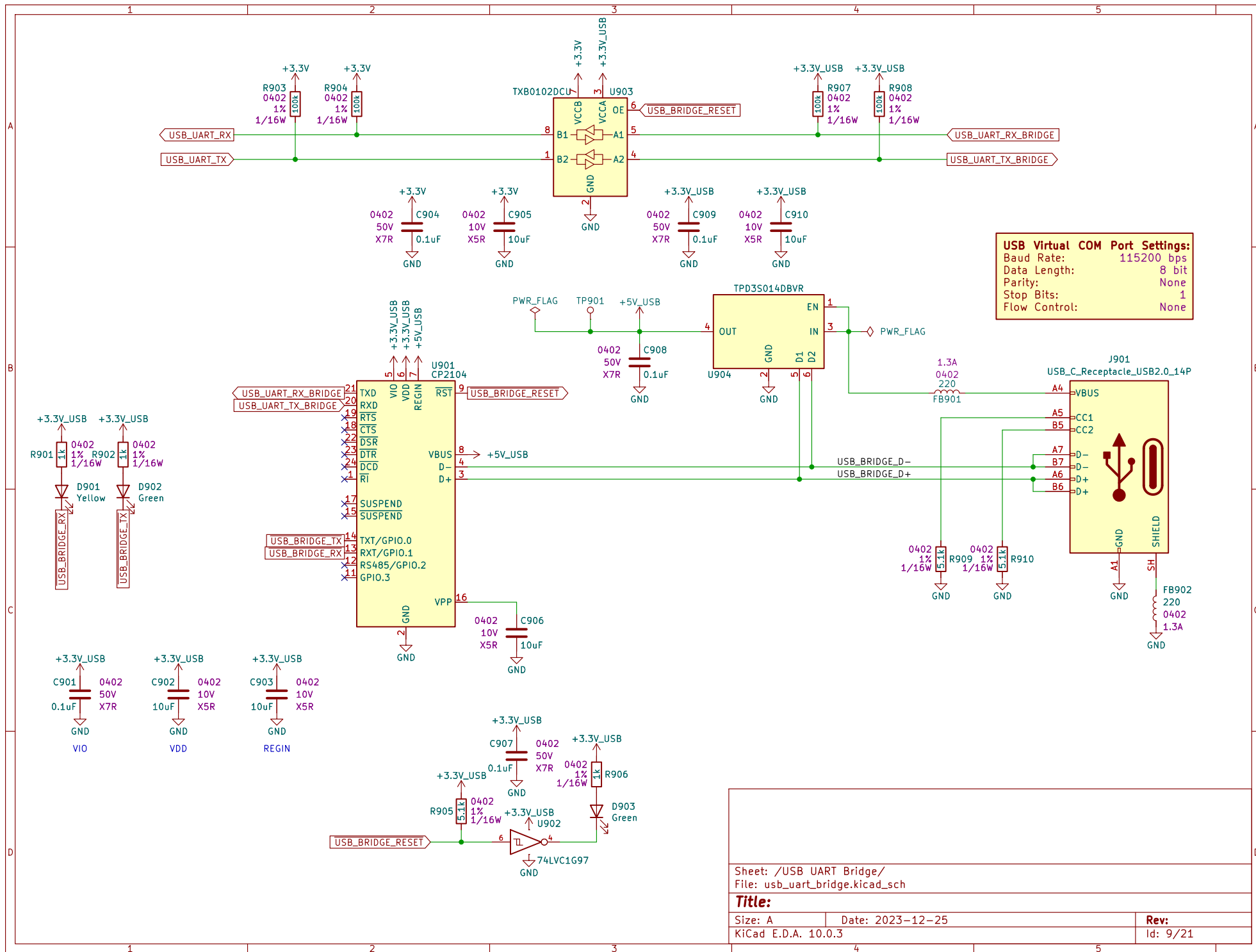
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch

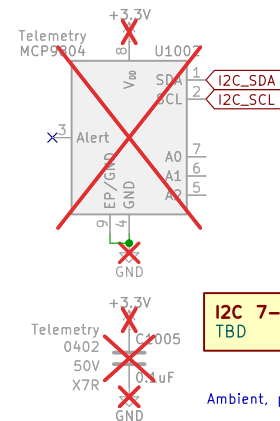
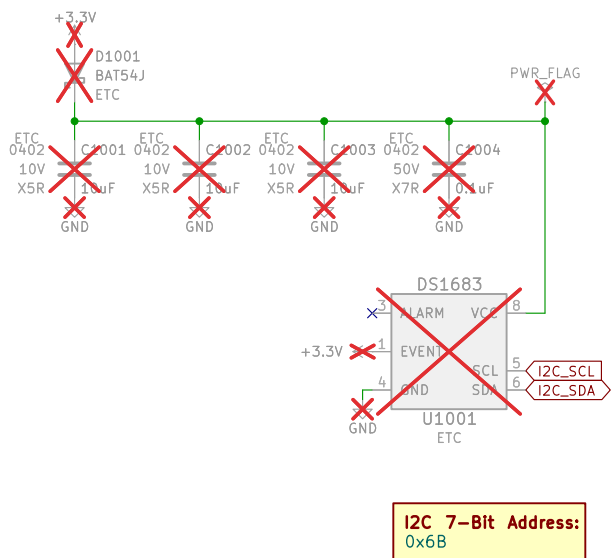
Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

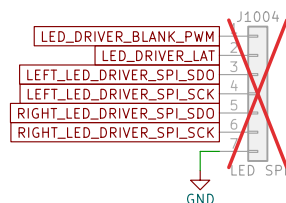
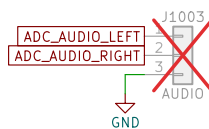
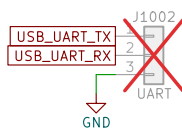
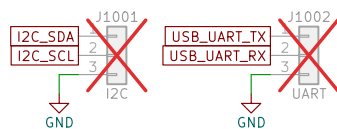
Rev:
Id: 6/21







Test Points

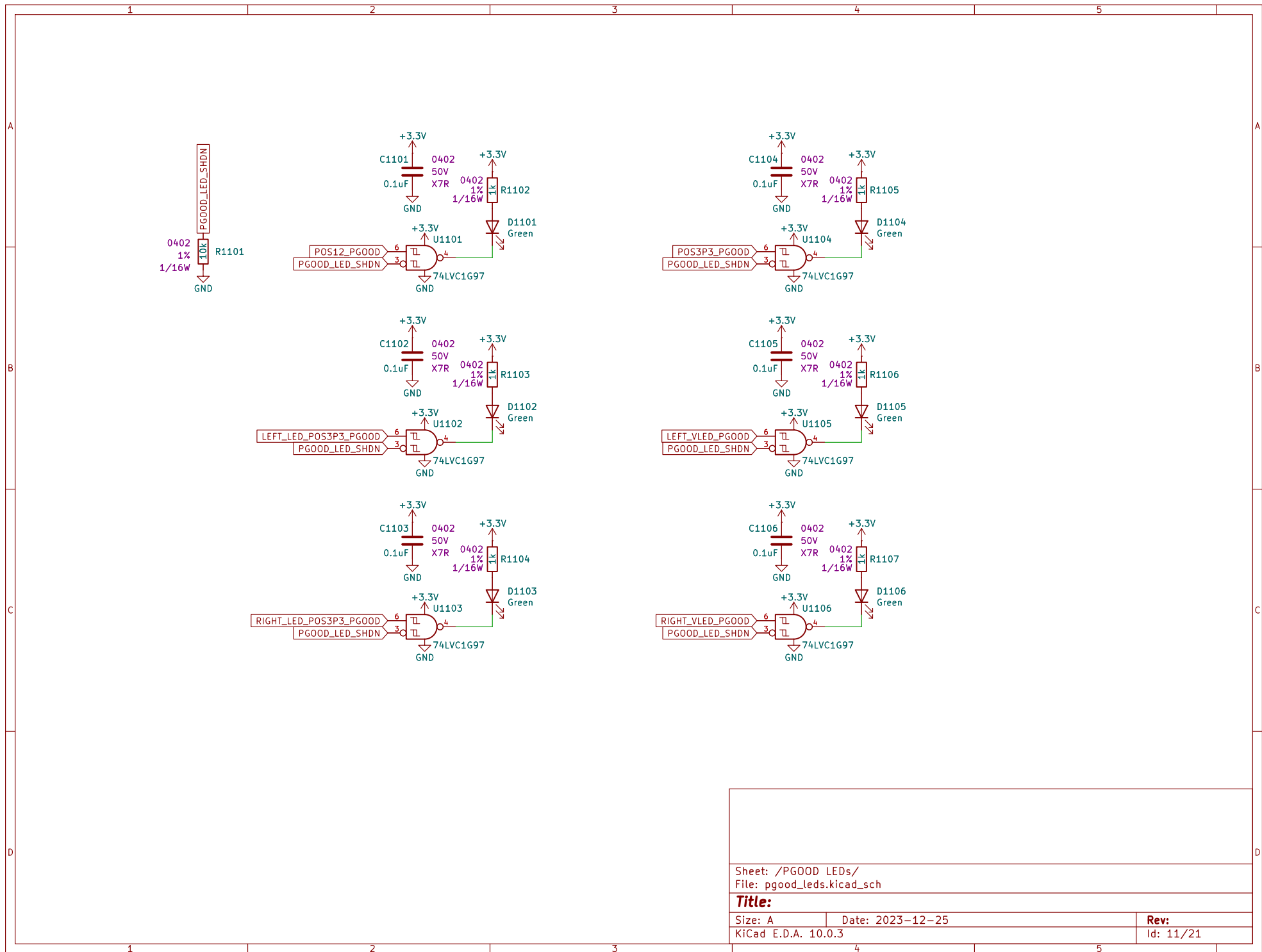


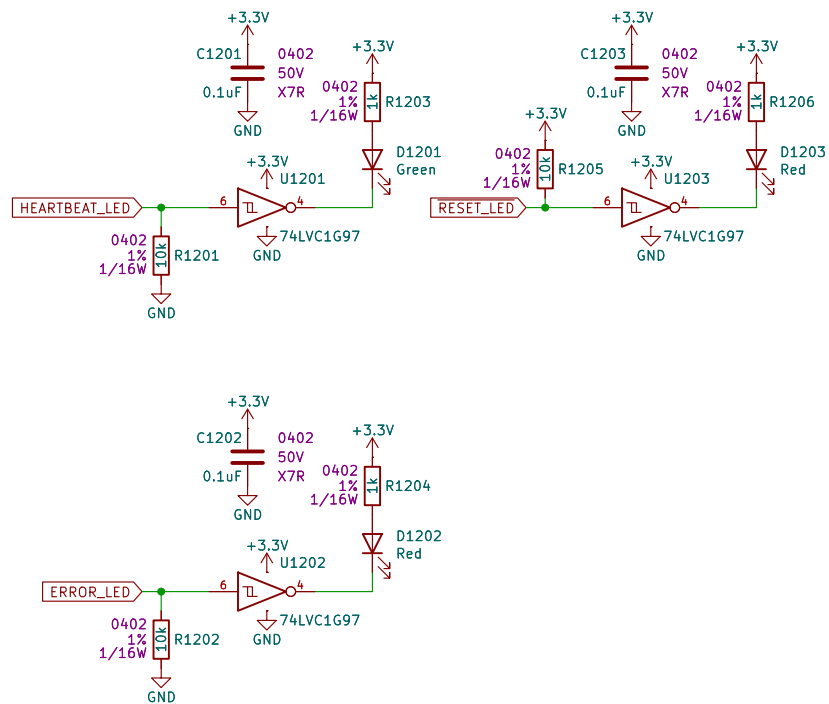
Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/21



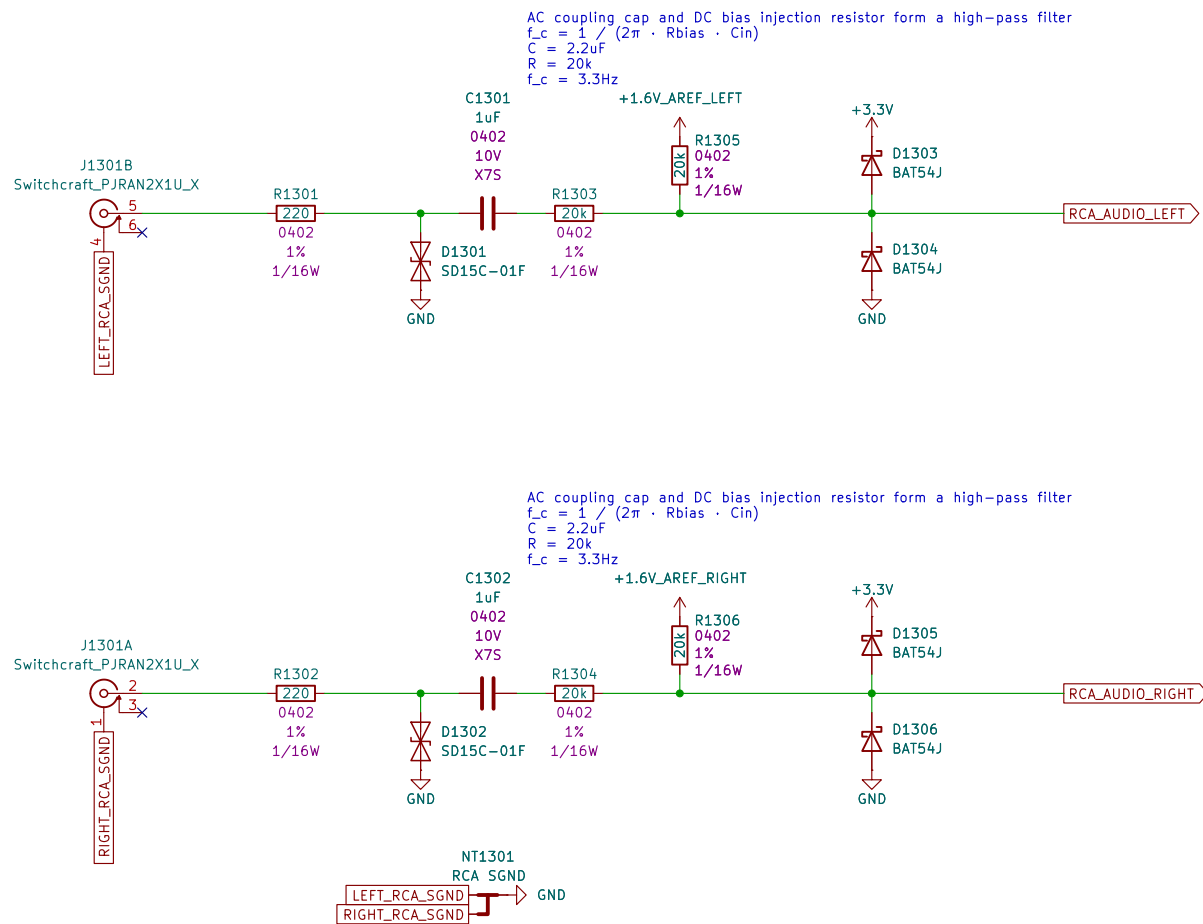


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/21

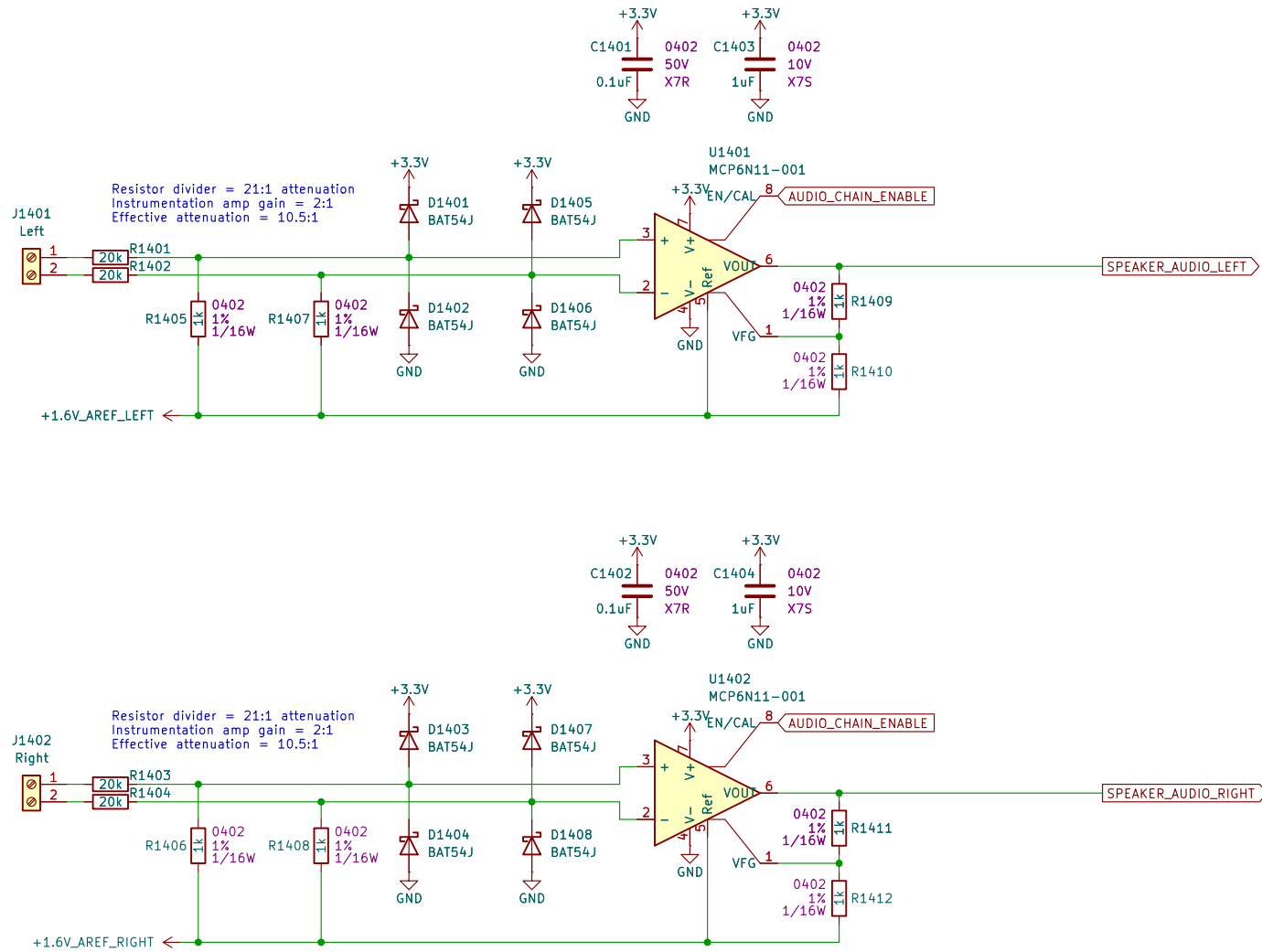


Sheet: /RCA Signal Conditioning/
 File: rca_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 13/21

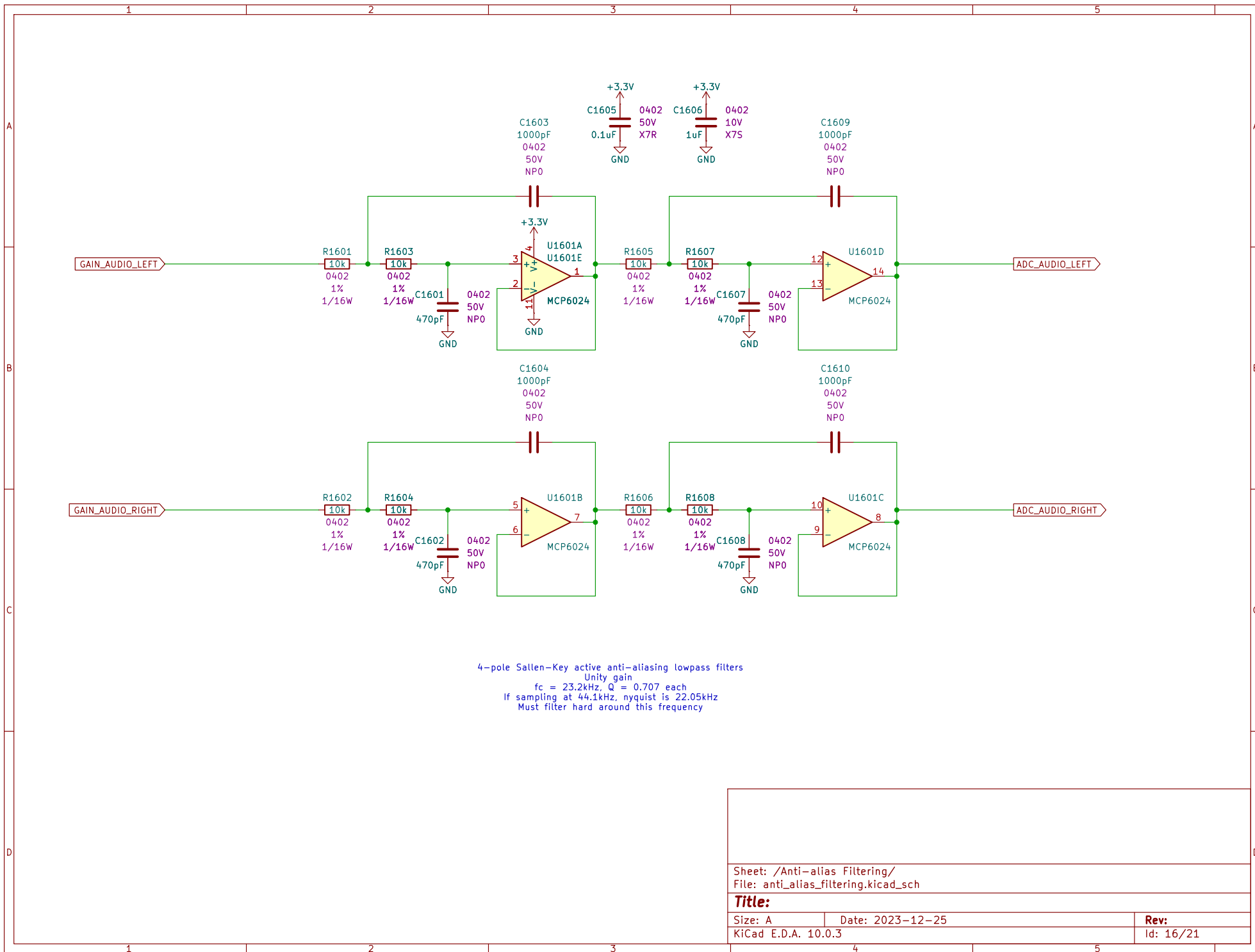


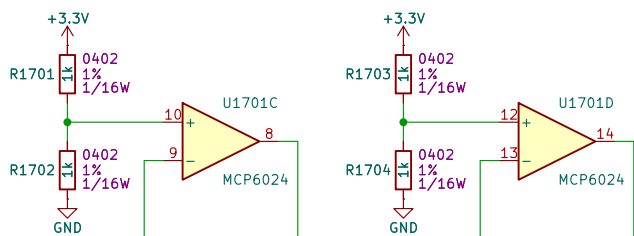
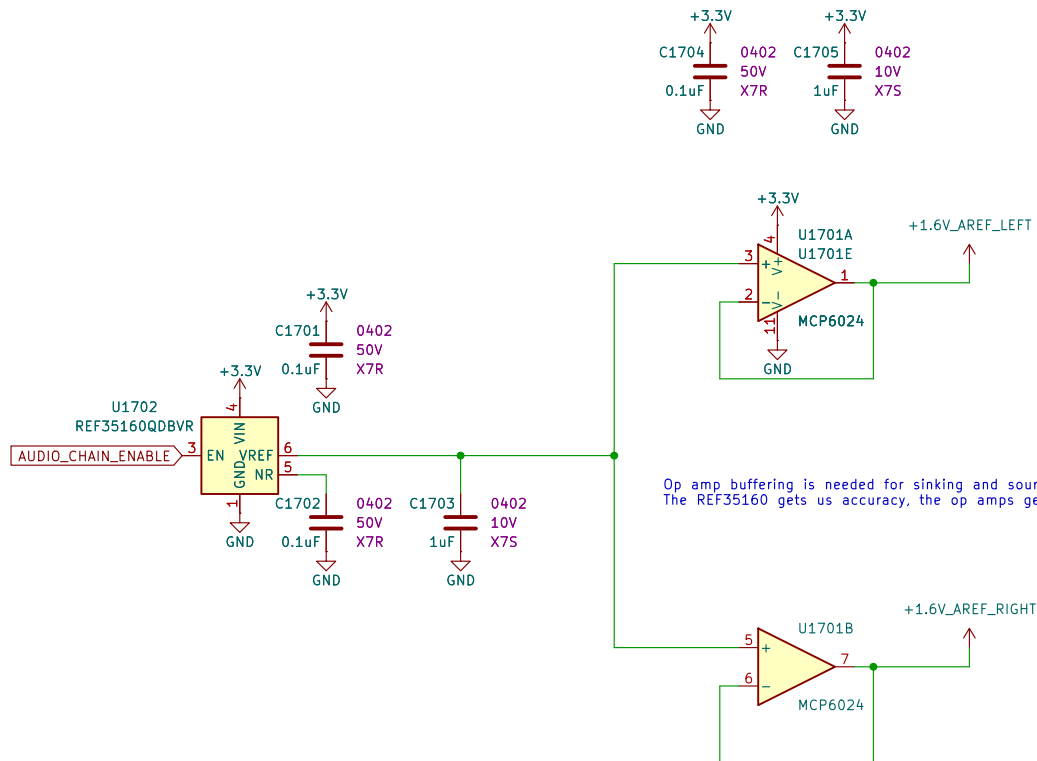
Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 14/21



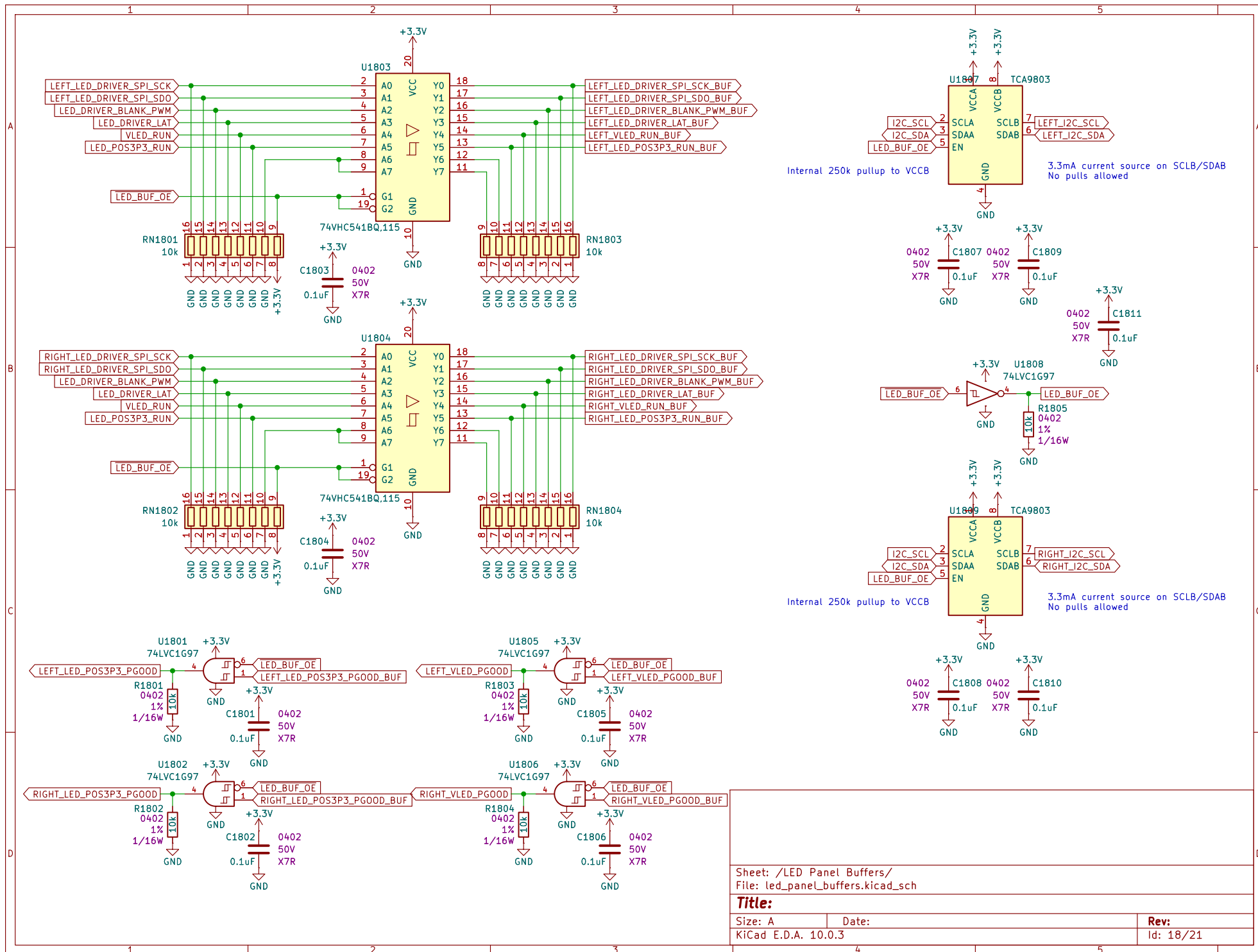


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/21



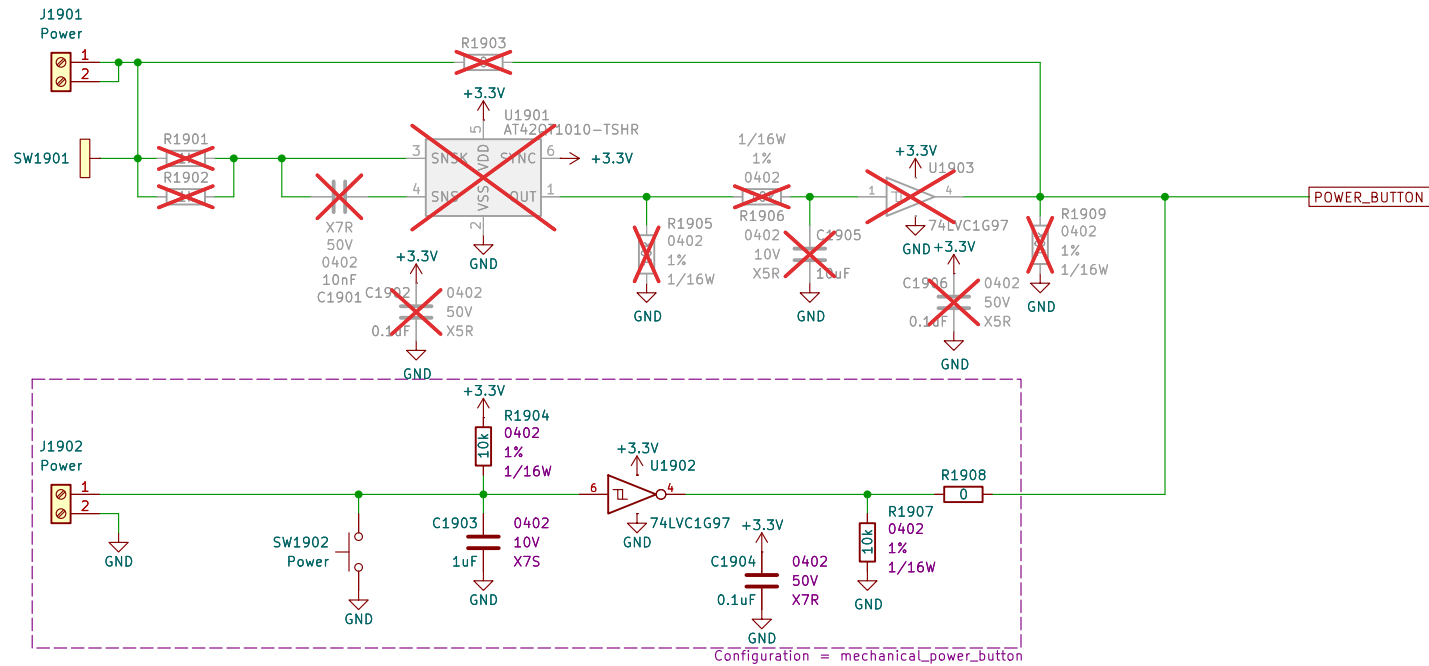
Sheet: /LED Panel Buffers/
 File: led_panel_buffers.kicad_sch

Title:

Size: A
 KiCad E.D.A. 10.0.3

Date:

Rev:
 Id: 18/21



N1901
Adafruit Rotary Encoder Module

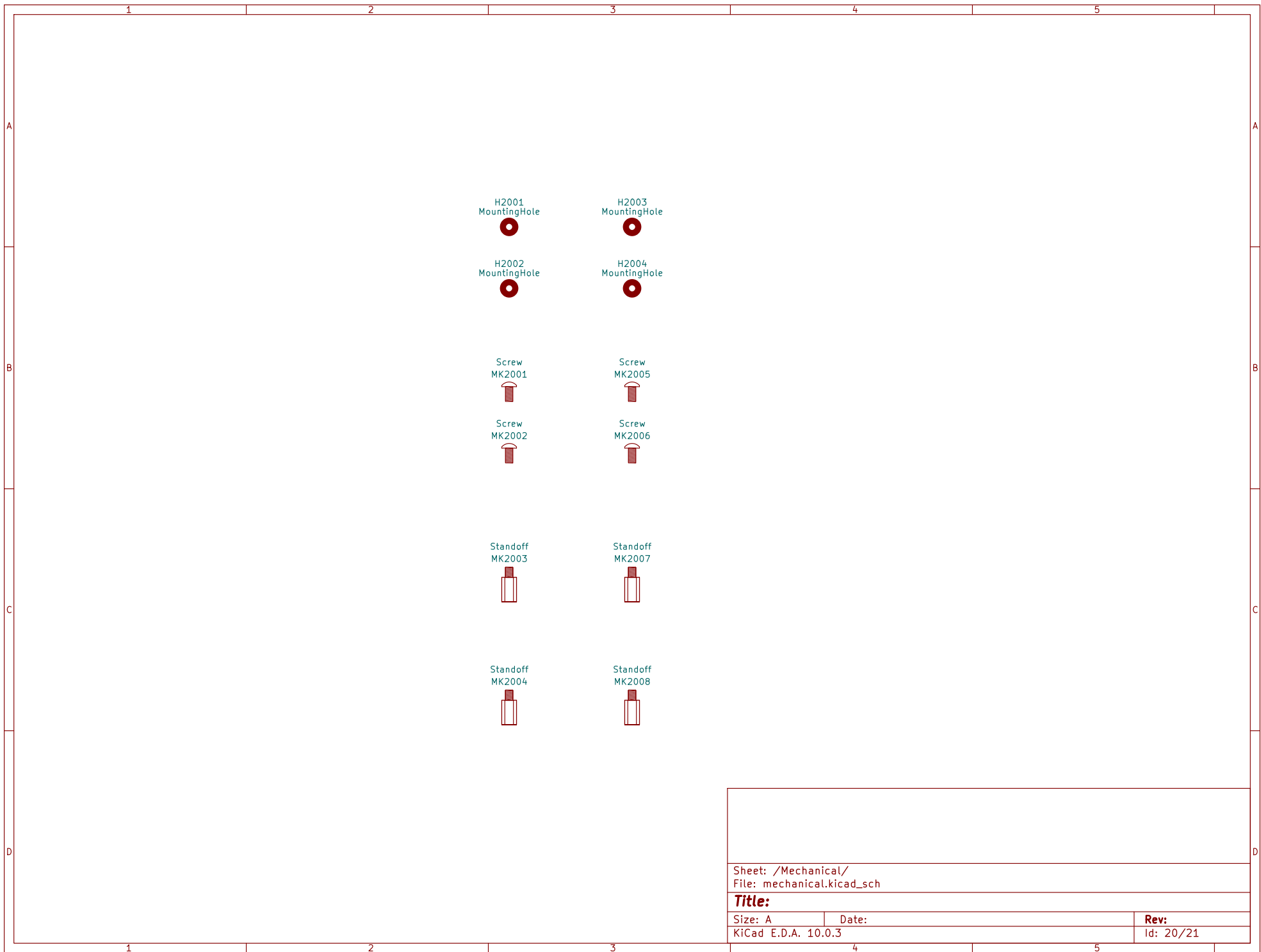
I2C 7-Bit Address:
0x36

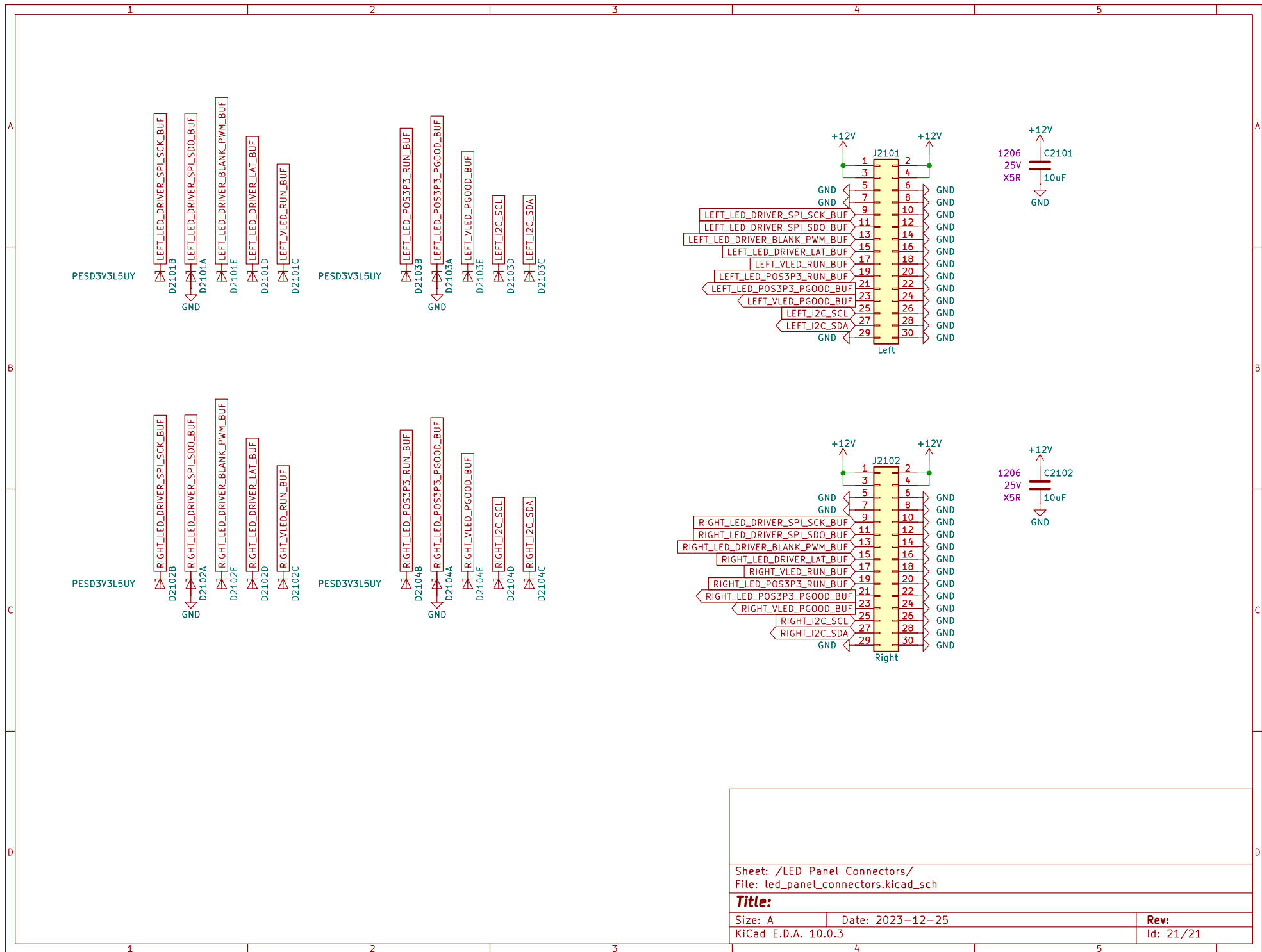
Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A Date:
KiCad E.D.A. 10.0.3

Rev:
Id: 19/21





Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 21/21